

Friday 16th January, 2022

Independent computers connected together to share data / resources

Set of independent computers interconnected to share data / exchange data or resources

Ques know the difference between Data and Information.



Components attached to systems to work as one can be:

- Computer
- Printers
- Routers

2. Link / path

could be physically joint
such as:-

④ Sending / receiving
sockets

Coaxial cable
Fibre Optics
Twisted pair

physical

Ques Define socket & its components

could be wireless:-

5 Protocol) The rules
of communication

radio

Internet

cellular

wifi

could be sent through messages

text

video

audio

Ques How computer Network works? (Dataflow)

• Packet Switching / Switching packet

Ques How are data broken into pieces and reassembled back again without losing data

Ques What is the technique behind packet switching

* Address — IP Address

↳ Internet Protocol Address

We are currently using IP 6

Ques know the background behind IP Address

Ques Difference between IP version 4 & IP version 6

5 classes
1, 2, 3, 4, 5

Ques What are the groups of IP version 4, Be able to identify the group of an IP address. (Know the Calculations) - Know the Mask, Host

Ques For data to move from one place to another, what do we need? Ans — IP Address

2 main items for data to move from one place to another
— Port Number

Ques Know the port number of each protocol

* Data transmission

— Synchronization

— Asynchronization

* Network types

PAN - Personal Area Network

LAN - Local Area Network

this modern Router switch

MAN - Metropolitan Area Network

WAN - wide area network

* Network Criteria

Que How can you measure Network Performance using ~~Delay~~

- Delay time

- Response time

- Throughput

How to measure

Network Performance

Que Formulae for Measuring ~~Performance~~

Que Robustness

→ Encrypting & Decrypting

What Julius Caesar did

So an unauthorized

person won't see

Ques 2×3 - If he wanted to send M O N E Y

M O N E Y

M in alphabet is 12

$$12 + 3 = 15 \rightarrow P$$

O in alphabet is 14

$$14 + 3 = 17 \rightarrow R$$

$$F(p) = p - 3 \rightarrow \text{what subtractions did}$$

Network Topology

- Bus - Ring - Bitonic
- Star - Butterfly - Mesh

Mesh

4	— 6 — 9	columns
1	1 1	1 4 2
8	— 2 — 4	9 6 4
1	1 1	18 7 8
1	— 18 — 7	Row again

The router will first sort a row

4 6 9	7 6 4
2 4 8	9 8 18
1 7 18	Again

Will then sort the column

1 4 8	1 2 4
2 6 9	1 6 4
4 7 18	8 9 18



Now, it's sorted

Will now sort the row again

1 4 8
9 6 2
18 7 4

Ques Techniques behind why each Network are applied

Network Architecture

- Peer to Peer
- client to server

Ques Write a program where client gets to communicate with server.

Communication Method

- Unicast
- Multicast
- Broadcast

Ques Mention the algorithms behind them

Protocol

OSI | TCP | IP

Ques Know each protocol for

layer	Physical layer	Lib	each layer
1	Physical layer	→ bit	
2	- Data link	→ Frame	
3	- Network	→ Packet	
4	- Transport	→ Segment	
5	- Session		
6	- Presentation		
7	- Application		

Friday 30th January, 2026

Transmission Media — Guided,
— Unguided,

Que What are the 2 types of transmission media

Routing Algorithms | Flooding, Shortest Path

Que How does TCP work? and how does Server & Client work in it

Que What are the algorithm and how does it work — Shortest path

Que Difference b/w STM and ATM

Que Dijkstra → how does it work?

Demonstrate encryption — which structure / type to use to use

What is UDP, TCP, port numbers etc, and how does it work

Download PyCharm Community-

Name your file (Server and client)

Create

Que- // We are trying to implement the communication of Client and Server.

→ This is the type of protocol
sock - stream (for TCP)

(Client & Server)
Before the 2 can communicate, they must use port number and IP address. And they must have the same port number to communicate.